



CAR $\gamma\delta$ T cells for cancer immunotherapy. Is the field more yellow than green?

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Abstract

Engineered immune cell therapy to treat malignancies refractory to conventional therapies is modernizing oncology. Although $\alpha\beta$ T cells are time-tested chassis for CAR, potential graft versus host disease (GvHD) apart from cytokine toxicity and antigen escape pose limitations to this approach. $\alpha\beta$ T cell malignancy challenges isolation and expansion of therapeutic T cells. Moreover, $\alpha\beta$ T cells may pose toxicity risk to inflammation sensitive vital tissues bearing the tumor. The HLA independent, multivalent, versatile and systemic anti-tumor immunity increases the desirability of $\gamma\delta$ T cells as an alternate chassis for CAR. Indeed, CD19 $\gamma\delta$ CAR T cell therapy to treat advanced lymphoma reached a milestone with the fast track status by FDA. However, reduced tumor-toxicity, homing, in vivo persistence and heterogeneity limits the translation of this therapy. The field is gaining momentum in recent years with optimization of gene delivery approaches and mechanistic insights into co-signaling requirements in $\gamma\delta$ T cells. There is a renewed interest in customizing design of CAR guided by the biology of the host immune cells. Progress has been made in the current good manufacturing practice compatible expansion and engineering protocols for the $\delta 1$ and $\delta 2$ T cells. $\gamma\delta$ CAR T cells may find its niche in the clinical situations wherein conventional CAR therapy is less suitable due to propensity for cytokine toxicity or off-tumor effect. As the therapy is moving towards clinical trials, this review chronicles the hitherto progress in the therapeutic engineering of $\gamma\delta$ T cells for cancer immunotherapy.

Keywords Gamma delta T cells · Chimeric antigen receptor · Cell therapy · Immuno-oncology

Abbreviations

aAPC	Antigen presenting cells
ADCC	Antibody dependant cellular toxicity
CCR	Chimeric Co-Stimulatory Receptors
CNS	Central nervous system
CRISPR/cas9	Clustered regularly interspaced short palindromic repeats
DAP10	DNAX—activating Protein 10 Endoplasmic reticulum kinase

FDA	Food and drug administration
GD2	Disialoganglioside
GMP	Good Manufacturing Practice
GvHD	Graft versus host disease
HLA	Human leucocyte antigen
IFN γ	Interferon gamma
IL-2	Interleukin -2
iPSC	Induced pluripotent stem cell
LAN1	A human neuroblastoma cell line
MART-1	Melanoma—associated antigen recognized by T cells
MCSP	Maternal and child survival program
MHC	Major histocompatibility complex
MICA/B	MHC class I chain-related protein A and B
mRNA	Messenger RNA
NK	Natural killer cells
NKG2D	Natural killer cell receptor
NKG2DL	Natural killer group 2D receptor and ligands
NSCAR	Non signalling chimeric antigen receptor

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NSG	NOD-SCID IL2R γ ^{null} mice
pERK	Type I transmembrane ER-resident protein kinase /protein kinase RNA-like
RORC	Register of registrable controllers
scFv	Single-chain variable fragment
TCR	T-cell receptor
ZA	Zoledronic acid

Introduction

$\gamma\delta$ T cells are innate lymphoid cells that detect infected or malignant host cells with broader molecular patterns of cellular stress [1]. The best described subsets involved in cancer immunosurveillance are the adaptive-like $\delta 1$ (+) and innate-like $\delta 2$ (+) cells. The $\nu\gamma 9 + \nu\delta 2$ subsets predominate the peripheral blood and recognize the phosphoantigens. Germ-line encoded activating (e.g., NKG2D) and inhibitory (e.g., NKG2A) functional receptors expressed in the $\gamma\delta$ T cells detects and lyse the target cells independent of HLA. Although the ligand of $\delta 1$ (+) TCR is not crystallized, the $\delta 1$ (+) T cells recognize the ligands of NKG2D, NKp30, and/or NKp44 in the tumor cells and effect endogenous cytotoxicity [2, 3]. Both the subsets got distinct chemokine, integrin expression pattern and unique tissular distribution [4]. In general, $\delta 1$ + cells persist longer with reduced exhaustion than $\delta 2$ + T cells in vivo. Majority of the clinical trials are performed with $\delta 2$ + T cells due to optimized expansion and scalability unlike $\delta 1$ + T cell. However, the mechanistic details of tumor antigen recognition in $\delta 1$ + T cells is not fully elucidated [5, 6].

The functional heterogeneity of $\gamma\delta$ T cells are further influenced by the tumor microenvironment as is reflected in the protumor and antitumor functions of the subsets [7–9]. The dichotomy of antitumor IFN γ + $\gamma\delta$ T cells and protumor IL17 + $\gamma\delta$ T cells is described in the preclinical and clinical settings. Foxp3 + $\gamma\delta$ T cells suppress the antitumor effector cells [10]. The cellular and molecular attributes and antitumor functions of $\gamma\delta$ T cells are detailed in the contemporary topical reviews [11–13].

The versatile immunity of $\gamma\delta$ T cells against multiple immunological stress makes this cell type an attractive candidate for adoptive immunotherapy of cancer. Experimental evidence suggests that $\gamma\delta$ T cells present antigens to the $\alpha\beta$ T cells contributing to systemic immunity [14]. Apart from the remarkable safety profile, $\gamma\delta 2$ T cells can be scaled up in vitro in clinically relevant numbers with zoledronic acid, making adoptive immunotherapy more viable [15].

Nonetheless, adoptive immunotherapy with $\gamma\delta$ T cells gave modest response rate possibly due to heterogeneity, reduced cytotoxicity, diminished cytokine secretion and limited in vivo persistence [16]. Importantly $\gamma\delta$ T cells lack the adaptive specificity and proliferative potential against

specified cognate antigen constraining its application in cellular immunotherapy. Certain virus dependent cancer (Burkitt's lymphoma due to Epstein Bar Virus infection) can make the tumor cells insensitive to $\gamma\delta$ T cell mediated lysis [17]. Although pre-treatment of tumor cells with ZA can make them more sensitive, genetic modification of $\gamma\delta$ T cells can multiply the overall therapeutic potential.

Engineering $\gamma\delta$ T cells with CAR: the chronology of progress

CAR Therapy is a topic that is extensively reviewed [18, 19]. Briefly, Chimeric Antigen Receptors (CARs) typically combine the antigen-binding domains of an antibody with TCR signalling domain (CD3 ζ) integrating co-stimulatory domains, so that the antigen recognition events are translated as TCR signalling; leading to the robust effector functions in an HLA independent manner [20]. Importantly, CARs recognize the antigens in an HLA independent manner to overcome tumor escape by HLA downregulation. However, conventional $\alpha\beta$ based CAR T cells suffer from the 'on-target-off tumor effect, cytokine release syndrome and the consequent neurotoxicity apart from the graft versus host disease (GvHD) in the allogeneic settings. Although the perpetuation of CAR T cells beyond tumor clearance guards future relapse, its persistence can be less desirable in vital tissues sensitive to high grade inflammation [21].

Switching the armature of CARs from $\alpha\beta$ to $\gamma\delta$ T cells can potentially resolve some of the caveats of conventional CAR Therapy. With the advancement and optimization of non-viral and viral gene transfer technology, there is a renewed interest in the field to engineer $\gamma\delta$ T cells to further improve the therapeutic benefits. Recent reports reveal augmented tumor-specific toxicity and safety of CAR $\gamma\delta$ T cells compared to native $\gamma\delta$ T cells against solid and liquid tumors [22]. CAR $\gamma\delta$ T cells are emerging as an appropriate choice to target malignancies of the T cells and central nervous system [23]. However, there are around a dozen of preclinical studies employing genetic redirection of $\gamma\delta$ T cells with CARs, even less so proceeding to clinical trials. In this context, the present review focusses on, the progress of genetic engineering in $\gamma\delta$ T cells with CARs and a projection of the future direction of CAR $\gamma\delta$ T cell therapy. The evolution in the preclinical CAR $\gamma\delta$ T cell therapy-briefly highlighting the context, the method, results and the take away message from these individual studies-are chronicled in the following subsections (Fig. 1).

CAR enhances the antitumor functions of $\gamma\delta$ T cells

Claudia Rossig's laboratory reported the first in vitro study on redirecting CARs against CD19 and GD2 expressed in

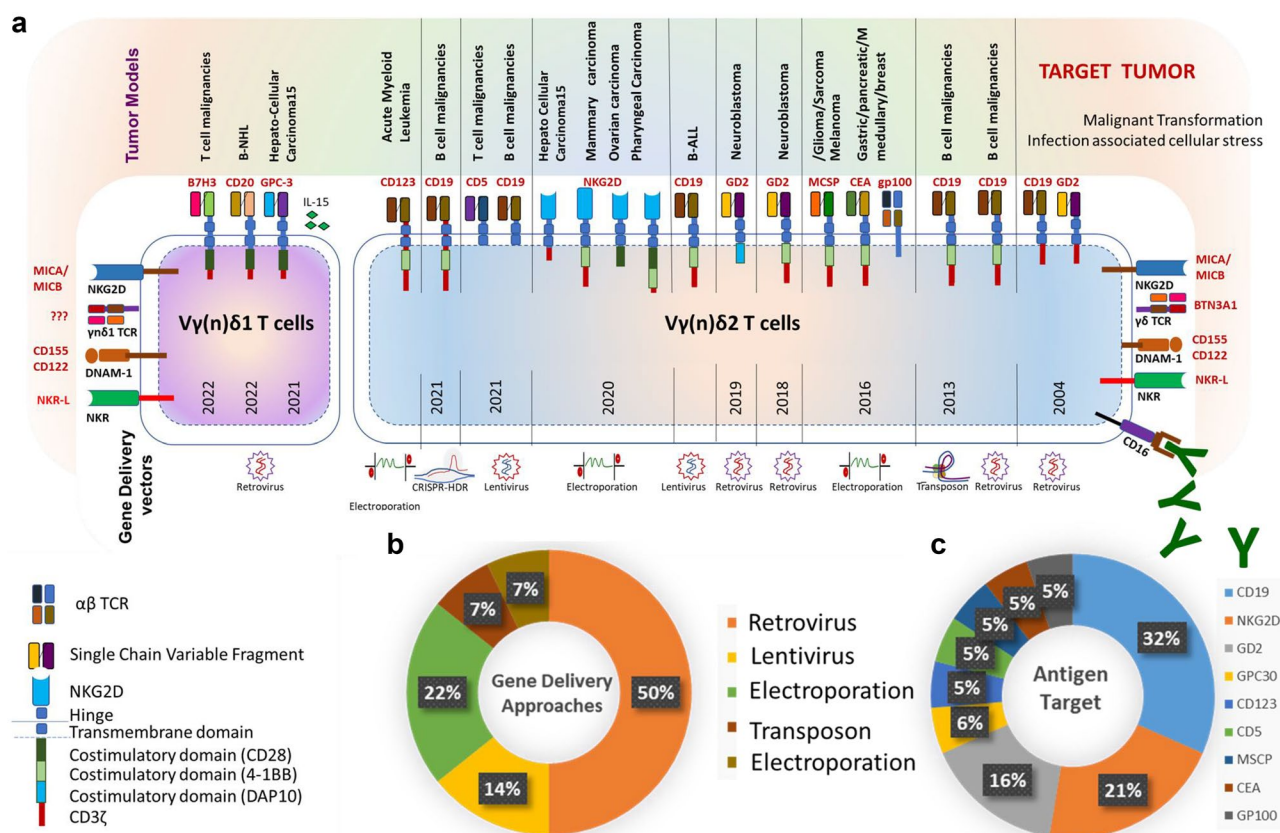


Fig. 1 Progress in targeting malignancies with CAR $\gamma\delta$ T cells

the clinically challenging malignancies of B cell and neuroectodermal lineage. The study compared the efficacy of first-generation CAR with single variable fragment (scFv) against CD19 and GD2 and signalling domain derived from CD3 ζ [24]. The CAR transduction efficacy was 73% with upregulation of proinflammatory markers such as CD69 and IFN γ . The anti CD19 CAR and anti GD2 CAR expressing $\gamma\delta$ T cells lysed the cell lines expressing the respective antigen with at least 2–threefold enhancement in cytotoxicity [24]. The effect was remarkable in the liquid tumor than solid tumors, possibly due to the peripheral blood origin of these tissue resident cells.

CAR $\gamma\delta$ T cells with polyclonal repertoire controls tumor cells in vivo

Five years later, the next report describing CAR $\gamma\delta$ T cell came from the laboratory of Laurence Cooper. Considering the polyclonality and need for mass production, Cooper's group engineered $\gamma\delta$ T cells from the peripheral blood with second generation CAR containing CD28 costimulatory domain [25]. The PBMC derived $\gamma\delta$ T cells were electroporated with Sleeping Beauty (SB) transposon/transposase system. The CD19 CAR $\gamma\delta$ T cells expanded to clinically

relevant proportions with > 30% transduction efficiency maintained up to 40 days. Because $\gamma\delta$ T cells needs to be infused multiple times, a protocol for expanding CAR $\gamma\delta$ T cells to very high numbers is critical for cGMP expansion of these cells. CD19 + artificial antigen presenting cells (aAPC) were used for expanding $\gamma\delta$ T cells to the tune of 10^9 which is remarkably higher than ZA + IL-2 with at least fivefold enhancement in the antigen specific lysis. A promising option to generate unlimited CAR $\gamma\delta$ T cells is to engineer T induced pluripotent stem cells (iPSCs) derived from T cells and reprogram them back into T cells. In vitro and in vivo studies indicated that the iPSC derived $\alpha\beta$ and $\gamma\delta$ like CAR T cells provided significant survival advantage to the human Burkitt lymphoma bearing NSG mice. This was the first extensive study that technically addressed the requirement for massively expanding the polyclonal CAR $\gamma\delta$ T cells in clinically relevant numbers.

mRNA-based CAR transduction for efficacy

Neils Schaft et al., in 2017 developed a safer strategy to overcome the 'on-target off-tumor' toxicities associated with CAR technology by introducing transient expression of tumor targeting receptors into $\gamma\delta$ T cells via mRNA

electroporation for the treatment of melanoma. The ZA expanded $\gamma\delta$ T cells were electroporated on day 10 with mRNA encoding either gp100 specific TCR or MCSP specific CAR (Second generation) and its expression peaked at the 24 h. The cytolytic capacity of the mRNA transfected cells studied by a 4 h Cr release assay in which ZA expanded $\gamma\delta$ T cells expressing gp100 Specific TCR lysed 30% of A375M cells (20:1). The MCSP CAR expressing $\gamma\delta$ T cells showed potent cytolytic activity (A375M: 40% at 20:1 and Mel526: 35% at 20:1). The engineered cells were further evaluated for innate cytotoxicity against cell lines not expressing MHC complexes in which the CAR expressing $\gamma\delta$ T cells exhibited 50% lysis when co-cultured [26]. Thus, the study demonstrated comparable cytolytic activity with $\gamma\delta$ CAR T cells to that of $\alpha\beta$ CAR T cells albeit with reduced cytokine production. Using safer non-integrating gene delivery approach this study demonstrated enhanced tumor sensing capability of $\gamma\delta$ CAR T cells in solid tumor system.

Co-stimulation-only chimeric antigen receptor (CCRs) to enhance specificity

Because CARs are designed against an antigen and not against tumor, ‘on-target off-tumor’ cytotoxicity remained as a major concern in the conventional CAR therapy. Dysregulated release of proinflammatory cytokine by the conventional CAR T cells needs refinement as persistence inflammatory CD8 + T cells in the CNS can cause neurotoxicity.

Conversely, antigen escape is a reported mechanism of reduced efficacy of CAR T cells [27]. Furthermore, shedding of the NKG2D and NKG2DL is associated with tumor evasion [28]. In this context, Anderson’s laboratory came up with a Chimeric Co-stimulatory Receptors (CCR) design to target neuroectodermal tumor antigen -disialoganglioside (GD2)- with cytoplasmic component being DAP10 which being the cytoplasmic adaptor of NKG2D [29]. In this orthogonal design, the primary signal is derived from the endogenous $\gamma\delta$ TCR whereas the secondary co-stimulatory signal is activated upon GD2 recognition. Unlike $\alpha\beta$ T cells, the tumor associated phosphoantigens induced activation dispense CD3 ζ domain in $\gamma\delta$ T cells. Consequently, the tumor lysis functions of the $\gamma\delta$ T cell is contingent upon encountering phosphoantigen and GD2 to augment ‘on-target-on tumor’ effect. The GD2 DAP10 CAR⁺ $\gamma\delta$ T cells lysed the GD positive cells but spared GD negative glioblastoma cell line—LAN1. The GD2-DAP10 CAR expressing $\gamma\delta$ T cells have comparable cytotoxicity to GD2-28 ζ against GD2 expressing LAN1 cell line. Furthermore, the CCR gives provision for broader antigen selection as antigens with tumor exclusivity are scarce. Perhaps, the split CAR approach is particularly relevant to CAR $\alpha\beta$ T cells which rely almost exclusively on HLA presented antigens.

Mechanistic basis of CAR signalling in $\gamma\delta$ T cells

Up until then, adequate consideration was not given to the mechanistic details of the CARs signalling in alternate host immune cell types such as $\gamma\delta$, NK and NKT cells. This often led to iteration of CAR structure originally designed for $\alpha\beta$ T cells in the preclinical studies. Myriads of ‘supra-physiological’ signalling during CAR T cell expansion eventuates sustained tonic signalling and exhaustion. Using single cell analysis by mass cytometry and network plasticity analysis, Anderson et al. noted that expansion did not increase the sensitivity to stimulation but heightened tonic signalling, reducing network plasticity with increased exhaustion signatures [30]. Because the tonic signalling was associated with CD3 ζ signalling, a series of CCRs against multiple tumor antigens with CCRs (GD2-CD28, CD33-CD28 and) were then retrovirally expressed in V δ 2 + cells [31]. GD2-CD28 did not trigger pAKT or pERK activation in line with reduced tonic signalling. Functionally, the receptor gave only a modest enhancement in cytotoxicity against GD2 positive LAN1 cell lines. Considering the paucity of antigen specific markers and shedding of NKG2D/NKG2DL in Acute Myeloid Leukemia model, the authors generated a CCR targeting CD33 with DAP10- the signalling partner of NKG2D [32]. CD33-DAP10 CAR antigen specifically lyse the tumor cells sparing the healthy myeloid cells indicating that the CCR strategy further enhance the ‘on-target on-tumor’ effect of CAR $\gamma\delta$ T cells. Recently, Shu wang’s group targeted AML by electroporating anti-CD123 CAR-encoding mRNA to $\gamma\delta$ 2 T cells with promising results in the preclinical settings [33]. In any case, there is increasing optimism that AML can be targeted by CAR $\gamma\delta$ T cells.

CAR $\gamma\delta$ T Cells to target glioblastoma: antigen presenting function is a bonus

Although the host immune cell choice for CARs widened, the biology and therapeutic function of CAR expressing immune effector cells were never systematically compared. As the field of conventional CAR therapy against glioblastoma was advancing, the clinical efficacy and safety profile of conventional CAR T cells targeted to GD2 antigen was still insufficient. Capsomidis et al., characterized the expansion potential, homing and antigen presenting functions of peripheral blood derived and retrovirally transduced GD2-28 ζ CAR expressing $\gamma\delta$ T cells. The authors included conA and ZA for expanding δ 1(+) and δ 2(+) subsets. δ 1(+) subsets are known to be more ‘adaptive’ than ‘innate’ phenotype with the participation of CAR T cells in the antibody dependant cellular toxicity (ADCC) and antigen presentation to $\alpha\beta$ T cells. δ 1 CAR T cells demonstrated more naïve and less exhausted phenotype whereas δ 2 CARs showed signs of exhaustions in terms of PD1 and TIM3 expression

even during expansion phase. The data shows that $\delta 1 +$ and $\delta 2 +$ subsets expressing GD2 CAR specifically proliferated and transmigrated towards multiple GD2 expressing neuroblastoma cells in vitro. Finally, the authors give evidence for the professional antigen presenting functions of CAR + V $\delta 2$ + T Cells presenting MART1 tumor antigen to MART1 TCR + $\alpha\beta$ T Cells [34]. The study characterized and compared GD2 CAR expressing $\delta 1$ and $\delta 2$ T cells subsets with that of the $\alpha\beta$ CAR T cells and revealed the multipotent functions of V $\delta 2$ + T Cells.

cGMP expansion of CAR $\gamma\delta$ T cells

cGMP protocol to manufacture $\gamma\delta$ CAR T cells in clinically relevant numbers is prerequisite for therapeutic infusion. Differentiating $\gamma\delta$ like T cells from CAR engineered iPSCs are previously demonstrated [35]. An alternative approach is the co-expansion of cytokine-induced killer cells (CIK) and $\gamma\delta$ T cells expressing CD19CAR [36]. Recently, Shu Wang's group communicated a protocol for largescale expansion (2000-fold) of V $\gamma 9$ V $\delta 2$ T cells [36]. Irradiated K562 aAPCs expressing CD64, CD86 and CD137L and anti-CD3 antibody OKT3 was used to co-culture PBMCs preactivated with ZA and IL-2. Combined with $\alpha\beta$ T cells depletion, 2 to 15 billion of functional V $\gamma 9$ V $\delta 2$ T cells were generated from 2 mL of peripheral blood [37].

Towards generating universal CAR $\gamma\delta$ T cells against solid tumors

With most of the $\gamma\delta$ CAR T cell studies restricted to hematological malignancies, Shu Wang's group explored the potential of NKG2D as the extracellular domain—owing to its expression in multiple solid tumors—along with combinations of co-stimulatory domain from CD28, 4-1BB and CD3 ζ [38]

NKG2D is a germline encoded activating receptor which engages tumor expressed ligands such as MICA/B to trigger cytokine production and cytotoxicity. The group also evaluated DAP12 with single ITAM motif instead of DAP10 as its cytoplasmic domain. It has been reported that multiple ITAM motifs in the case of DAP10 are associated with increased toxicity in murine studies [39]. With electroporation as the transient mode of gene expression, the efficacy was above 90% and viability greater than 60%. NKG2D-CD3 ζ first generation CAR showed the highest in vitro antitumor toxicity, cytokine production and the authors also optimised the cryopreservation conditions for 'off-the-shelf therapy'. Multiple infusion of CAR $\gamma\delta$ T cells every 3 days with ZA pre-treatment enhanced the persistence of $\gamma\delta$ T cells and tumor clearance in the NSG model of colorectal and ovarian cancer bringing CAR $\gamma\delta$ T cells closer to clinical translation.

Non signalling CAR: signalling domain may be dispensable in CARs designed for $\gamma\delta$ T cells

T cell therapy for T cell malignancies is paradoxical owing to the mutual lysis of T cells sharing targeted antigen, T cell aplasia of therapeutic T cells [40]. Spencer's group explored the possibility of an artificial receptor without signalling domain named as non-signalling CAR (NSCAR) to target CD5 and CD19 against T-ALL and B-ALL, respectively [23]. NSCARs and its cognate antigens are downregulated in T cell co-culture, demonstrating receptor shedding thereby correlating to transduction efficiency but not reducing the viability of co-expanding cells. NSCAR expression in $\alpha\beta$ T cells did not alter the antitumor function. Owing to small size, multiple small CAR expression potentially prevents tumor antigen escape. Intriguingly, in $\alpha\beta$ T cells NSCAR did not alter the antitumor functions, revealing the dispensable role of co-stimulatory signals for $\gamma\delta$ T cell antitumor function requiring further investigations. This milestone manuscript casts light into the co-stimulatory and cytolytic mechanisms of $\gamma\delta$ T cells. The study calls for reconsidering the requirements for co-stimulatory domains to tailor CARs for $\gamma\delta$ T cells [41].

$\gamma\delta$ CAR T cells benchmarked against conventional $\alpha\beta$ based CAR T cells

Besser's laboratory generated CAR T cells from peripheral blood derived $\alpha\beta$ and $\gamma\delta$ T cells and made systematic comparison of the antitumor functions in vitro and in vivo [22]. The purity of $\gamma\delta$ T cells lentivirally transduced with conventional after $\alpha\beta$ T cell depletion was near 100%. The priming of tumors with ZA augmented the cytotoxicity and cytokine production in both $\alpha\beta$ and $\gamma\delta$ CAR T cells in vitro against CD19 (+) and CD19 (–) negative NALM-6 cells indicating its ability to target antigen escape variants even in allogeneic settings. Multiple infusions of $\gamma\delta$ CAR T cells in the NSG bearing NALM6 cell lines pre-treated with ZA achieved comparable tumor control to that of $\alpha\beta$ CD19 CAR T cells. Detailed mechanistic insights at the molecular level will further illuminate engineering strategies to optimize $\alpha\beta$ and $\gamma\delta$ CAR T cell therapy designs.

An end-to-end process to generate iPSC derived CRISPR/Cas9 engineered CD19 CAR $\gamma\delta$ T cells

Zoledronic acid exposure of the $\gamma\delta$ T cell results in inter-donor heterogeneity of the product due to differential drug sensitivity [42]. Century Therapeutics reported the optimization of research grade and clinical grade $\gamma\delta$ CAR-iT lines by reprogramming the $\gamma\delta$ T cell derived iPSCs. $\gamma\delta$ CAR-iT lines can be genetically modified to expand functional $\gamma\delta$ CAR-iT cells. $\gamma\delta$ CAR-T cells were redirected by CRISPR

gene editing, which were subsequently re-differentiated into $\gamma\delta$ CD19CAR T cells. $\gamma\delta$ CAR-iT cells targeting CD19 was found to lyse the NALM-6 cells in vitro and in vivo. The single dose of $\gamma\delta$ CAR-iT cells reduced >95% of the tumor burden. This innovative project can potentially alter the speed and reach of $\gamma\delta$ CAR-T cells therapy translation [43].

Expanding off-the-shelf $\delta 1$ + T cells to target oncofetal proteins in solid tumors

Although much attention was given to the preclinical trials on $\delta 2$ + T cells, lack of scalable protocol side-tracked the therapeutic exploration of $\delta 1$ + T cells. Lack of a cGMP grade protocol for expanding and engineering $\delta 1$ + T cells restricting research groups to pursue a low hanging fruit, in this case the $\delta 2$ + T cells. Stimulating the PBMC with antiCD3 antibody in presence of IL-15 after depleting $\alpha\beta$ T cells and CD56 + NK cells was found to drastically improve the proliferation and expansion of $\delta 1$ + T cells. Barisa's group at University Colleges of London (UCL) demonstrate that this protocol also permits retroviral transduction of B7H3-CAR to target T cell malignancies [44]. Recently, Adicet Therapeutics reported an innovative platform for expansion of V $\delta 1$ (+) gamma delta T cells targeting oncofetal proteins with constitutive IL-15 expression and demonstrated their antitumor efficacy in solid tumor -hepatocellular carcinoma [45]. This integrated approach for the first time demonstrated massive proliferation rate (2000-fold) with enhanced persistence for the anti-glypican-3 (GPC-3) CAR + $\delta 1$ T cells. Instead of targeting tumor associated antigen, this group explored GPC-3 -the oncofetal protein-which are long been investigated owing to its tumor exclusive re-expression [46]. This study for the first time targeted an oncofetal protein rather than the conventional tumor associated antigen.

Off-the-shelf V $\delta 1$ T cells targeting CD20: moving on to clinics

Accessibility of the CAR therapy to ineligible patients is an issue that needs to be addressed at the preclinical and clinical levels. Interestingly, the V $\delta 1$ T cells retrovirally transduced with CD20 CAR -an antigen targeted by Rituximab for the treatment of r/r B-cell NHL- was found to have CAR dependent and independent antitumor functions without eliciting the graft versus xeno-host disease. The CD20 CAR + (ADI-001) $\delta 1$ T cells were found to elicit both innate and adaptive immune response, suggesting the impact of these cells in the systemic immunity. The naïve like ADI-001 exhibited a robust antitumor effector function in vitro and in vivo. However, unlike the FDA approved and affordable ZA, expansion of $\delta 1$ T cells by antagonistic antibody

needs to be refined to enhance the accessibility of $\delta 1$ CAR T cells therapy [47].

Progress in translating engineered $\gamma\delta$ T cells to clinical trials

The field is now moving forward to translate the preclinical investigations into clinical trials. Several multinational pharmaceuticals are involved in generating novel formulations of CAR $\gamma\delta$ T cells across variety of indications [48]. The CD19, CD7 and NKG2D CAR expressing $\gamma\delta$ T cells are registered for clinical trial. Importantly, ADI-001 is being tested in a dose escalation phase I study against late-stage r/r NHL. The results of the first engineered, allogeneic off-the-shelf CD20 CAR $\delta 1$ T cell therapy (GLEAN-1 trial) were recently released by Adicet Bio. There were no side effects due to—GvHD or high-grade cytokine release or neurotoxicity—in the reporting period. The overall response rate of the study was 75% and complete response rate was 50%. The V $\delta 1$ T cell research reached a critical milestone with the fast-track designation by FDA [49].

The details of the clinical studies are summarised in the Table 1.

Challenges of CAR $\gamma\delta$ T cell therapy

Although promising, CAR $\gamma\delta$ T cell therapy needs to overcome multiple biological, manufacturing and clinical hurdles. First of all, the impact of CAR over expression in the biology of $\gamma\delta 1$ + and $\gamma\delta 2$ + T cells needs to be established. $\gamma\delta$ T cells got higher sensitivity to activation induced cell death (AICD) than $\alpha\beta$ T cells, impacting the choice of gene delivery approaches therapeutic administration [31]. Furthermore, majority of the gene transfer techniques used retroviral or electroporation approaches for delivering CARs to $\gamma\delta$ T cells (Fig. 1b), (Table 2). An optimized protocol judiciously combining the gene delivery approaches in terms of timing, frequency and dose is paramount for basic and clinical studies. Whether the CAR overexpression leads to enhanced IPP in the $\gamma\delta$ T cells itself and making them susceptible to mutual cell lysis is a possibility that needs to be experimentally addressed. $\gamma\delta$ T cells needs to be established. Detailed mechanistic understanding of gene delivery in $\alpha\beta$ T cells and $\gamma\delta$ T cells may advance the cGMP protocol. While the preclinical results were encouraging, reduced persistence of $\gamma\delta$ T cells in vivo leads to residual tumor growth demanding multiple infusions of $\gamma\delta$ CAR T cells incrementing the host immunogenicity to CAR [51]. Therefore, humanizing the receptor is crucial. The influence of inter-donor variations in TCR chain usage, T cell differentiation pattern and cytokine requirement for expanding $\gamma\delta$ CAR T cells needs to

Table 1 CAR $\gamma\delta$ T cell therapy: clinical trials

Sl. No	Year	$\gamma\delta$ T cell subset	Transgene	Cancer type	Dosage/frequency	Phase/stage	NCT identifier	Academia/company
1	2017	$\delta 2(+)$	CD19 CAR	B-Cell lymphoma, acute lymphoblastic leukemia and chronic lymphoblastic leukemia	Unknown	Phase I/Not yet recruiting	NCT02656147	Beijing Doiing Biomedical Co., Ltd. Singapore-Malasia
2	2019	$\delta 2(+)$	NKG2DL CAR	Colorectal cancer triple negative breast cancer sarcoma nasopharyngeal carcinoma prostate cancer gastric cancer	3×10^8 , 1×10^9 , 3×10^9 (Thrice)	Phase I/Not yet recruiting	NCT04107142	CytoMed Therapeutics Pte Ltd, China
3	2021	$\delta 2(+)$	CD7 CAR	CD7 + T cell malignancies	$0.2\text{--}5 \times 10^6$ / kg (Once)	Phase I/Not yet recruiting	NCT04702841	PersonGen BioTherapeutics (Suzhou) Co., Ltd., China
4	2019	$\delta 2(+)$	MGMT (With drug temozolomide (TMZ))	Glioblastoma multiforme (GBM)	–	Phase I/recruiting	NCT04165941	University of Alabama at Birmingham IN8bio, Inc, United States
5	2021	$\delta 2(+)$		Patients With Relapsed AML After Transplantation	–	Phase I/recruiting	NCT04796441	Hebei Senlang Biotechnology Inc., Ltd. China
6	2021/22	$\delta 1(+)$	CD20	R/R B cell lymphoma	Unknown	Phase I/recruiting	NCT04735471	Adicet Bio, Inc, United States

be characterized. Although tissue specific residency reduces off-tumor toxicity, differential in vivo homing of CAR $\gamma\delta$ T cells moderates the universality of the approach. Conversely, it remains undetermined as to how $\gamma\delta$ CAR T cells overcome physical and immunological homing barriers in solid tumors [52]. The molecular insights on the functions of $\gamma\delta$ CAR T cells in the immunosuppressive, hypoxic and metabolically competitive tumor microenvironment would shape the development in solid tumor therapy. $\gamma\delta$ T cells specific signalling events through the cytoplasmic domain needs to be compared in the CARs and its endogenous receptor, before deciding its suitability in the CAR design. Assessing the high density immune phenotype of the $\gamma\delta$ CAR T cells as a function of pro and antitumor response would reveal additional therapeutic targets. Since the co-stimulatory requirements for $\gamma\delta$ T cells are unique, the immune check points targeting strategy for CAR $\gamma\delta$ T cells may be realigned.

So far there is no formal study comparing the toxicities of $\alpha\beta$, $\gamma\delta$ and NK cells based CAR T cells. An off-tumor effect due to recognition of phospho-antigen positive stressed healthy cells is a possibility. Although CAR significantly enhanced proinflammatory cytokine production, preclinical studies do not demonstrate a potential for grade 3 or above CRS [22]. Since $\gamma\delta$ T cells got immunosuppressive regulatory T cells, it is possible that tumor growth may resume due to potential protumor effect of total $\gamma\delta$ T cells [53]. Long-term clinical side effects of $\gamma\delta$ CAR T cells needs to be carefully investigated since the $\gamma\delta$ T cells are implied in multiple autoimmune diseases [54].

Conclusion and future directions

In the past decade, there is a commendable advancement in the cGMP level expansion of CAR $\gamma\delta$ T cells. In the context of the results from NKG2D and NSCAR study, the discretionary use of costimulatory domain in the CAR architecture needs to be guided by signalling and specific effector outcomes in the candidate effector cells. The promise of CAR $\gamma\delta$ T cells may be overstated in the absence of more convincing clinical data. Unlike the conventional CAR T cell space, the impact of the advancements in the gene delivery, expansion, tumor antigen discovery, cancer biology and CAR design; is less pronounced in the CAR $\gamma\delta$ T cells field. CRISPR/cas9 mediated genome editing guided by mechanistic studies combined with the developments in the iPSC technology may generate effective, safe and affordable CAR $\gamma\delta$ T cells. The field may be turning yellow but yes, collective clinical research progress in the academia and industry is required to harvest the therapeutic benefits of CAR $\gamma\delta$ T cells.

Table 2 Approaches for delivering CARs to $\gamma\delta$ T cells

Vector/attributes	Retrovirus	Lentivirus	Transposon	mRNA electrofection
Molecule	ss RNA	ss RNA	DNA fragment	Receptor-encoding mRNA
Cargo size	Upto 8 kb	Upto 8 kb	Upto 2 kb (size > 2 kb, reduces uptake efficiency)	< 4 Kb
Integration to the genome	Only in mitotically active cells	Integrates in both dividing and non-dividing cells	Integrates randomly in the host chromosome	negligible integration
Gene transfer efficiency	73 $\pm$ 12% [24] 58.6 $\pm$ 3.7% [29] 40–60% [34]	80% [23] 65.3 $\pm$ 18.3% [22]	30.7 $\pm$ 23.3% [25]	> 80% [26] 81% [36] 87.8% [50] 96% [37]
Stability of transgene expression	3 weeks	Upto 2 weeks	36 days	Few hours to 7 days
Vector induced apoptosis	High	High	Depends on the method of delivery	Very high
safety	Less safe (immunogenic)	Less safe (less immunogenic)	Safe (But random integration can be mutagenic)	Safe (high voltage can damage tissues)
Logistics of vector preparation	5 days	5 days	1–2 days	2 days
cost	Expensive	Expensive	Lower than viral vectors	Lower than viral vectors

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Code availability Not applicable.

Declarations

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Consent to participate Not applicable.

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